

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Theory of Computation

COURSE CODE: CSA13 01

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Design Task

Code of Conduct:

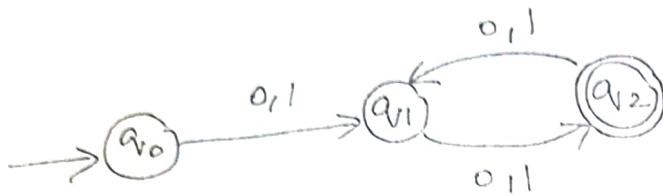
I Deepak V-19241021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

SIMATS ENGINEERING ASSESSMENT TOOL

1. Design the regular expression for the language represented by the given automata



2. Consider the following grammar

$S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|e$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

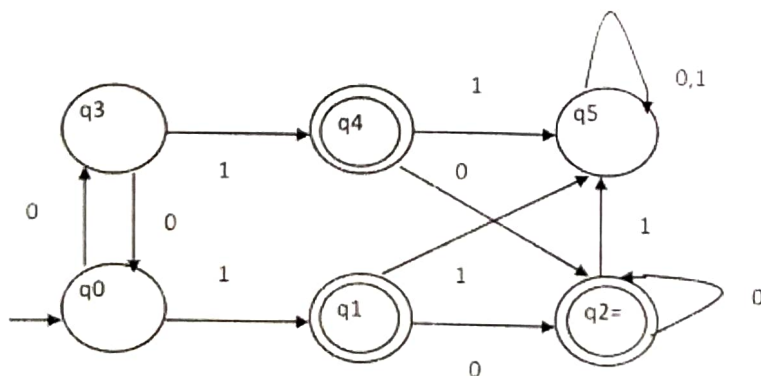
3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

00^*1+11^*0

4. i) Show that if L_1 and L_2 are regular then (L_1+L_2) also regular

ii) Is the language $\{a^p | p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



Regular Expression for the Given Automata

From the automaton, the first 2 symbols can either be 0 or 1 and automaton returns between q_1 and q_2 for every additional symbol

Transition Table:

State \ IP	0	1
$\rightarrow q_0$	q_1	q_1
q_1	q_2^*	q_2^*
q_2^*	q_1	q_1

$\rightarrow q_0$ is the initial state

$\rightarrow q_2$ is final state

Regular Expression:

$$RE = \underbrace{(0+1)(0+1)}_{\text{for}}^* (0+1)(0+1)(0+1)(0+1)^*$$

examples: $RE = (0+1)(0+1)^+$

$\rightarrow 01 \checkmark$

$\rightarrow 101 \times$

$\rightarrow 00 \checkmark$

$\rightarrow 0 \times$

NFA 20
near RE)

Queen,

$$S \rightarrow aS \mid T \mid U \mid \epsilon$$

$T \rightarrow b T c \mid \epsilon$

$$U \rightarrow aUb \mid \epsilon$$

A) Language definition:

- T generates $b^n x^n$
- U generates $a^n b^n$
- S add matching a and x

$$L = \{a^i b^j c^{i+j} \mid i, j \geq 0\} \cup \{a^{i+j} b^j c^i \mid i, j \geq 0\}$$

B) Derivation Tree for $aaabbb$:

$S \rightarrow aSu$; $S \rightarrow \cancel{U}u$, $U \rightarrow aUu$

$S \rightarrow aVb, V \rightarrow aVb$

S → add block, U → $\emptyset$

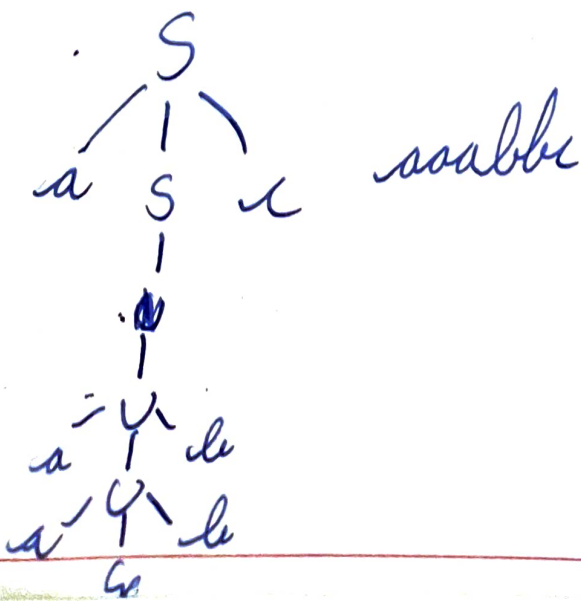
C) Ambiguous:

$$S =) \zeta$$

$$S \Rightarrow T \Rightarrow \zeta$$

$$S \Rightarrow V \Rightarrow \zeta$$

∴ Grammar is Ambiguous

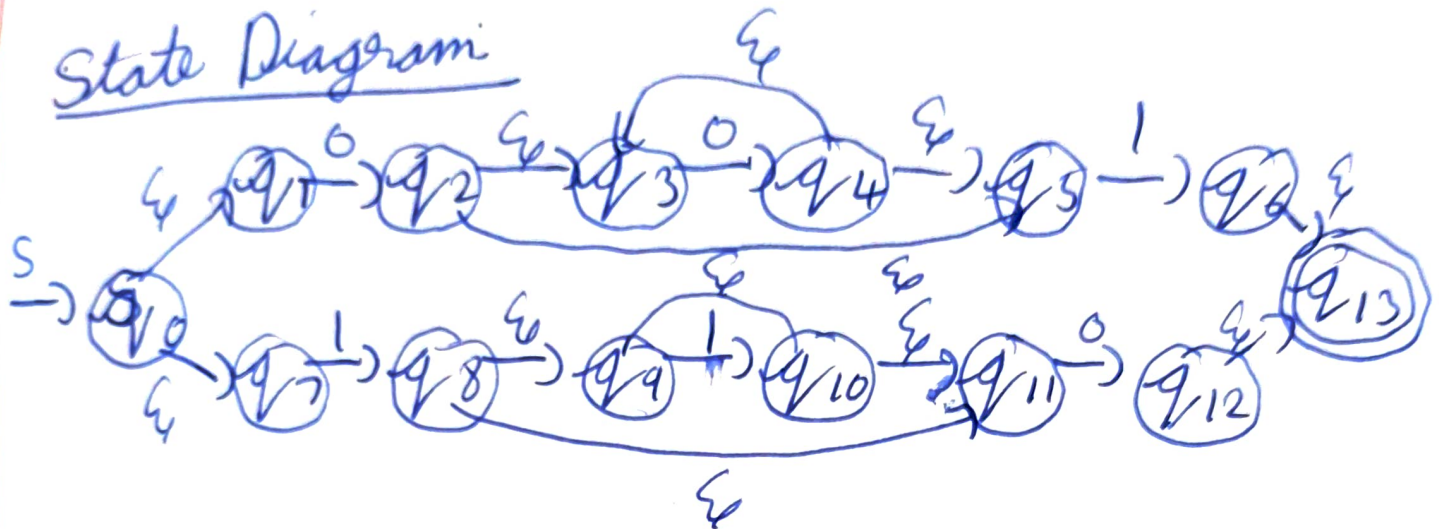


NFA for $00^*1 + 11^*0$

union RE,

$00^*1 + 11^*0$

State Diagram



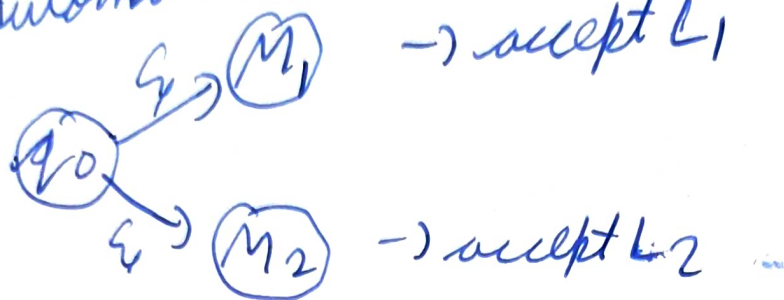
Transition Table:

State \ IP	0	1	ϵ
$\rightarrow q_0$	—	—	$\{q_1, q_2\}$
q_1	q_2	—	—
q_2	—	—	$\{q_3, q_5\}$
q_3	q_4	—	—
q_4	—	—	q_3
q_5	—	q_6	—
q_6	—	—	q_{13}^*
q_7	q_8 —	q_8	—
q_8	—	—	q_9
q_9	—	q_{10}	—
q_{10}	—	—	q_{11}
q_{11}	q_{12}	—	—
q_{12}	—	—	q_{13}^*
q_{13}^*	—	—	—

4) Regular Languages

1) Show that if L_1 and L_2 regular, $L_1 \cup L_2$ is regular.

Since L_1 and L_2 are regular, there exist finite Automata M_1 and M_2 accepting them. Construct



New NFA accepts: $L_1 \cup L_2$

$\therefore L_1 \cup L_2$ is regular

2) Is $L = \{a^p \mid p \text{ is prime}\}$ regular?

No, the language is not regular.

Assume L is regular and n be its pumping length.

Choose prime $p > n$

$$w = a^p$$

By Pumping Lemma, let

$$w = xy^k$$

where, $y = a^k, k > 0$

pump y $p+1$ times: $xy^{p+1} = a^{p+pk}$

$\therefore \{a^p \mid p \text{ is prime}\}$ not regular. $p + pk = p(1+k)$, $p \neq p(1+k)$ is composite.

DFA Minimization:

From the given DFA, final states are:
 q_1, q_2, q_4

non-final:
 q_0, q_3, q_5

→ No dead states

$P_0: \{q_1, q_2, q_4\} \{q_0, q_3, q_5\}$

State Transition table:

State \ IP	0	1
→ q_0	q_3	q_1^*
q_1^*	q_2^*	q_5
q_2^*	q_2^*	q_5
q_3	q_0	q_4
q_4^*	q_2^*	q_5
q_5	q_5	q_5

~~q_6~~

~~P_2 Input 0,~~

~~$P_2: \{q_1\} \{q_2\}$~~

$P_0: \{q_0, q_3, q_5\} \{q_1, q_2, q_4\}$

$P_1: \{q_5\} \{q_0, q_3\} \{q_1, q_2, q_4\}$

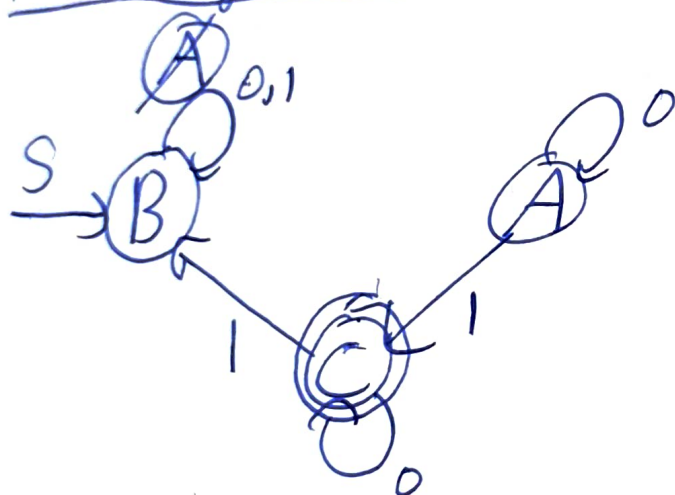
$P_2: \{q_5\} \{q_0, q_3\} \{q_1, q_2, q_4\}$

$A = q_5$

$B = q_0, q_3$

$C = q_1, q_2, q_4$

Minimized DFA:



SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

84/100